Quiz 2: Algorithm Analysis & Recurrences

SCS1308 Foundations of Algorithms

Question 1

For T(n) = 2T(n/2) + n, which substitution step correctly tests the hypothesis T(n) ≤ cn log n?

a. $T(n) \leq cn^2 \log n + n$

b. $T(n) \leq 2cn \log n + n$

c. $T(n) \leq c(n/2)\log n + n$

d. $T(n) \leq cn \log(n/2) + n$

e. $T(n) \leq 2c(n/2)\log(n/2) + n$ — CORRECT

Question 2

Consider binary search on a sorted array. Which is the most appropriate loop invariant for the search interval [low, high]?

a. The target is never outside the array

b. The target, if present, is always within the current search interval — CORRECT

c. The current interval always contains all elements of the array

d. The target is always equal to A[low]

e. low and high are always equal

Question 3

Using the Master Theorem, what is the tight bound for $T(n) = 2T(n/2) + n^2$?

a. $\Theta(n \log n)$

b. $\Theta(n^2)$ — CORRECT

c. $\Theta(2^n)$

d. $\Theta(n^2 \log n)$

e. $\Theta(n)$

Question 4

For T(n) = T(n−1) + 1, T(0) = 1, what does the recursion tree show?

a. A tree with n leaves

b. A single chain of n recursive calls plus the base case — CORRECT

c. A tree with 2^n leaves

d. A constant-depth tree

e. A binary tree with depth log n

Question 5

For $T(n) = T(n-1) + 2n$, $T(0) = 0$, which expression correctly represents the result after k iterations?

a. $T(n) = T(n-k) + k^2$

b. $T(n) = T(n-k) + 2k$

c. $T(n) = T(n-k) + 2^k$

d. $T(n) = T(n-k) + 2nk$

e. $T(n) = T(n-k) + 2kn - k^2 + k$ CORRECT

Question 6

Using the Master Theorem, what is the tight bound for $T(n) = 4T(n/2) + n^2$?

a. $\Theta(n \log n)$

b. $\Theta(n)$

c. $\Theta(n^2 \log n)$ CORRECT

d. $\Theta(n^2)$

e. $\Theta(2^n)$

Question 7

For $T(n) = 2T(n/2) + n$, a recursion tree has how many levels before reaching subproblem size 1?

a. 2^n

b. $\log_2 n$ CORRECT

c. $n/2$

d. 1

e. n

Question 8

Which substitution hypothesis is most appropriate for proving the upper bound $T(n) = O(n \log n)$ for $T(n) = 2T(n/2) + n$?

a. Assume $T(n) \leq cn$

b. Assume $T(n) \leq c \log n$

c. Assume $T(n) \leq cn^2$ and conclude $\Theta(n \log n)$

d. Assume $T(n) \leq 2^n$ and conclude $\Theta(n \log n)$

e. Assume $T(n) \leq cn \log n$ CORRECT

Question 9

Using the iteration method, what is the tight asymptotic bound of $T(n) = T(n/2) + 1$, with $T(1) = 1$?

a. $\Theta(n)$

b. $\Theta(2^n)$

c. $\Theta(\log n)$ CORRECT

d. $\Theta(1)$

e. $\Theta(n \log n)$

Question 10

A recursion tree for T(n) = 2T(n/2) + n is formed. What is the total non-recursive work at each level?

a. It increases by a factor of 2 at every level

b. It decreases by a factor of 2 at every level

c. It is always Θ(1)

d. It remains Θ(n) at every level CORRECT

e. It is always Θ(log n)

Question 11

Using the Master Theorem, what is the tight bound for T(n) = 3T(n/2) + n?

a. Θ(n)

b. $\Theta(n^2)$

c. $\Theta(3^n)$

d. Θ(n^(log₂ 3)) CORRECT

e. Θ(n log n)

Question 12

Consider T(n) = 2T(n−1) + 1, with T(0) = 1. Which is the tight asymptotic bound obtained by iteration?

a. Θ(log n)

b. Θ(n)

c. $\Theta(3^n)$

d. Θ(n log n)

e. $\Theta(2^n)$ CORRECT

Question 13

Using the iteration method, what is the tight asymptotic bound of T(n) = T(n−1) + n, with T(1) = 1?

a. Θ(n)

b. Θ(n log n)

c. $\Theta(2^n)$

d. Θ(log n)

e. $\Theta(n^2)$ CORRECT

Question 14

Consider the loop: for i = 1 to n, sum = sum + i. Which is the strongest suitable loop invariant immediately before iteration i?

a. sum contains the sum of all integers from 1 through n

b. i = n

c. sum = i

d. sum = i^2

e. sum contains the sum of all integers from 1 through i−1 CORRECT

Question 15

In proving a loop invariant, which statement establishes the initialization condition?

a. The invariant is true only after termination

b. The loop condition must be false before the first iteration

c. The invariant is true before the first iteration begins CORRECT

d. The invariant must be false after every iteration

e. The loop must execute exactly n times